

Persistent overtime use among fixed-term staff of Servicio de Salud Metropolitano Central (SSMC), Chile

Data preparation and exploratory analysis · C1

Group 01 · Alessandro Lavezzi · José Saavedra

Instructor: Tomás Fontecilla Correa · September 29, 2026

The problem and the decision we support

When overtime is paid almost every month, it is no longer occasional

Team proposal

Problem: overtime paid almost every month to the same workers

Population: fixed-term (contrata) staff of the SSMC, September 2024 – August 2026

User: SSMC human resources management units

Decision: which staff categories and positions to review first for staffing

Alternatives: keep paying overtime · create or expand positions · hire replacements · reassign staff

Why it matters

CLP 2,086 M

Between +25% and +50% overtime cost

overtime paid over 18 months in a single public hospital: Hospital San José de Victoria (Comptroller General, 2026)

the law allows overtime only for tasks that cannot be postponed, compensated first with time off; paid overtime with a premium (25–50%) is the exception (Law 18,834, arts. 66, 68–69)

Estrategia Nacional de Salud 2030: identifies “high territorial inequity in the distribution of human resources in the network” but sets no target for overtime (MINSAL, 2022, p. 29; own translation).

From W1 to C1: why we chose this line

Two candidates explored in W1 with the same data source

Candidate A

FOCUS

Payments above the legal limit of 40 daytime overtime hours per month (Law 19,104, art. 9)

W1 SIGNAL

182 person-months above the limit; 99 in a single month (January 2025)

ISSUE

The result depends on one month with atypical records

Candidate B (selected)

FOCUS

People who receive overtime in almost every month

W1 SIGNAL

364 of 1,776 people (20.5%) with overtime in $\geq 75\%$ of their months

STRENGTH

Covers 1 in 5 people and barely changes without January 2025 (20.5% $\rightarrow$ 19.9%)

What changed in C1

- Question reframed from “staffing gap” to “persistent use”: no staffing data available
- KPIs defined as ratios: numerator, denominator and time window
- Suspicious cases flagged, not deleted, with KPIs computed with and without them (19.0–20.5%)
- Persistence threshold (75%)
- Job titles unified (95 $\rightarrow$ 80 values) to compare across positions

Why B

- Broad, stable signal
- Less sensitive to recording errors: measures whether there was overtime across many months, not how much in one
- Uses all 24 months: better base for ML

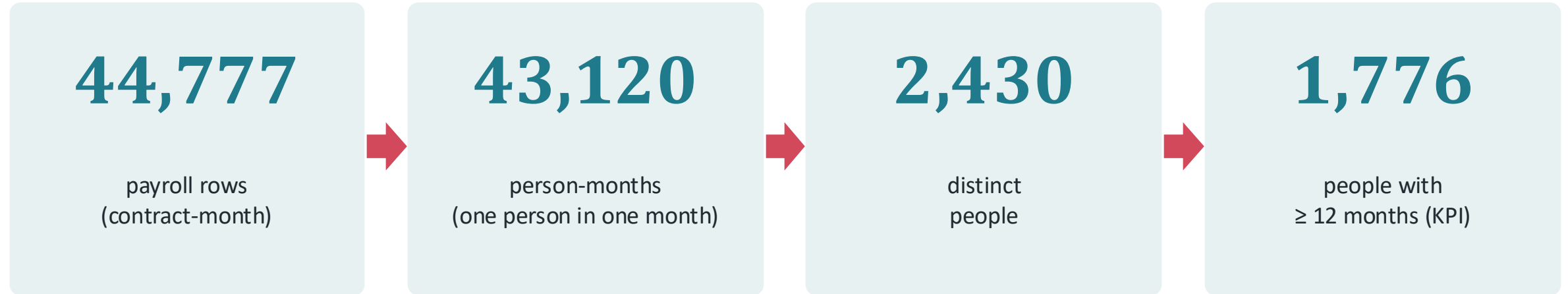
The seven steps

User, decision and success criteria are a team proposal

Step	In this project	Gap found
1. Decision	Where to review staffing first , where the user is the SSMC human resources units	No hiring costs or authorized staffing
2. Success criteria	Desired: less recurring overtime · Measurable today: persistence, frequency and pay share (baseline)	No official target
3. Business questions	Which staff categories and positions show persistent use, and how much of their pay does it represent?	The “why” cannot be answered
4. Business process	Authorization → work → attendance → payment → publication	We only see the payment
5. Entities, events and states	Person, contract, month · Event: overtime payment · State: persistent or not	No facility or unit
6. Data representation	Row = contract-month · Key = (month, name) · Fixed-term staff only	No national ID; text formats
7. Measurement	Persistence, frequency and pay share (slide 7)	Measures payments, not hours worked

Data and unit of observation

Fixed-term staff payroll, Active Transparency portal (AO006): 24 monthly files



- A payroll row is a contract in a month: 57–68 people per month have more than one contract
- So each person's contracts are summed within each month: one person contributes at most 24 person-months
- Check: total hours, amounts and pay are identical before and after aggregating
- No national ID (RUT): people are identified by name (risk of namesakes)

Glossary

Person-month: one person present in one month's payroll, with or without overtime.

Staff category (estamento): auxiliary, administrative, technical, professional or physician.

Overtime: daytime + night hours paid in the month.

Three KPIs, defined as ratios

Full-period values; there is no official target, so they serve as a baseline

20.5%

Persistence rate

People with overtime in $\geq 75\%$ of their months / people present ≥ 12 months

Purpose: identifies who uses overtime routinely, a signal of where to review staffing

Main KPI · 364 of 1,776

27.4%

Frequency

Person-months with overtime / person-months

Purpose: measures how common overtime is in a month and allows month-by-month tracking

11,805 of 43,120

14.6%

Share of gross pay

Median of (overtime amount / gross pay) among person-months with overtime

Purpose: measures how much pay depends on overtime (cost)

Excludes pay = 0

Example: a person with 24 months on the payroll and overtime in 18 of them (75%) is persistent. Overtime = daytime + night hours (holiday hours have no records). The KPI measures persistent use; it does not prove understaffing.

Quality: flag instead of delete

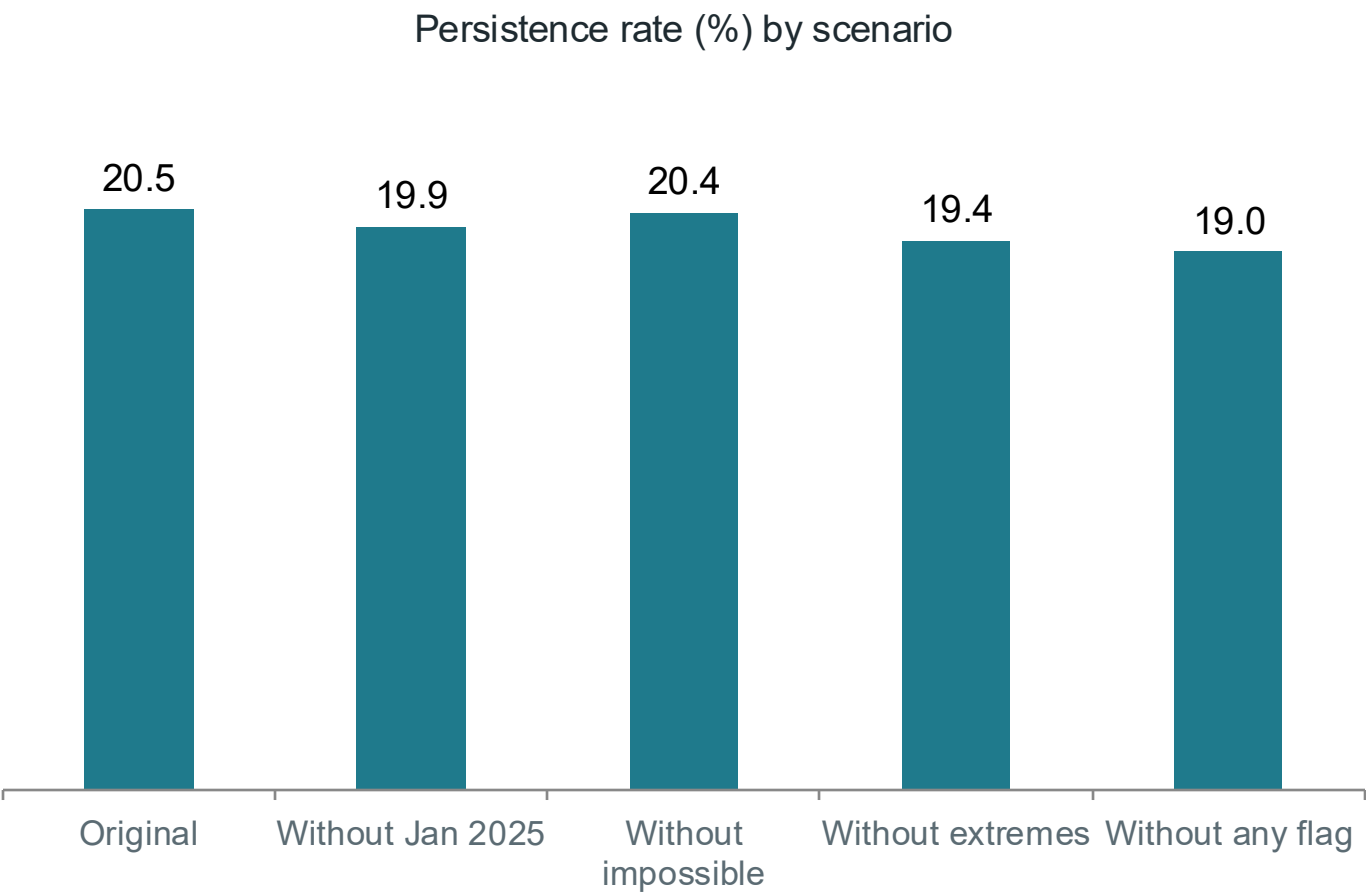
An outlier flag is not a deletion rule (Class 02)

Finding	Size	Decision
Missing values written as "-" (not blank)	1,825 in additional pay (all of January 2025); 3,065 in bonuses	Missing, not imputed; not used in KPIs
"No tiene" (none) in overtime	33,821 daytime and 36,551 night records	Recorded zero (no payment), not missing
Exact duplicates	27 copies (54 rows), 11 physicians, no overtime	Kept: they do not change the KPIs
Atypical month: January 2025	1,756 person-months; 99 of 182 cases above 40 daytime hours	Flagged
Staff category misrecorded: Dec. 2025	218 changes and 213 back	Most frequent category per person
Impossible values	9 person-months with overtime > gross pay	Flagged
Extreme values ($Q3 + 1.5 \cdot IQR = 157.5$ h)	486 person-months; 61% auxiliary staff	Flagged, not deleted

Flag = mark the case with a control column, without deleting it, to measure its effect (next slide). Five Cs: Clean and Consistent reviewed above; Conformed does not apply because no other sources are joined; Current: data up to Aug 2026; Comprehensive: covers fixed-term staff only.

KPIs barely change when flagged cases are excluded

The conclusion does not depend on the quality issues



19.0–20.5%

persistence across the five scenarios

26.5–27.4%

frequency

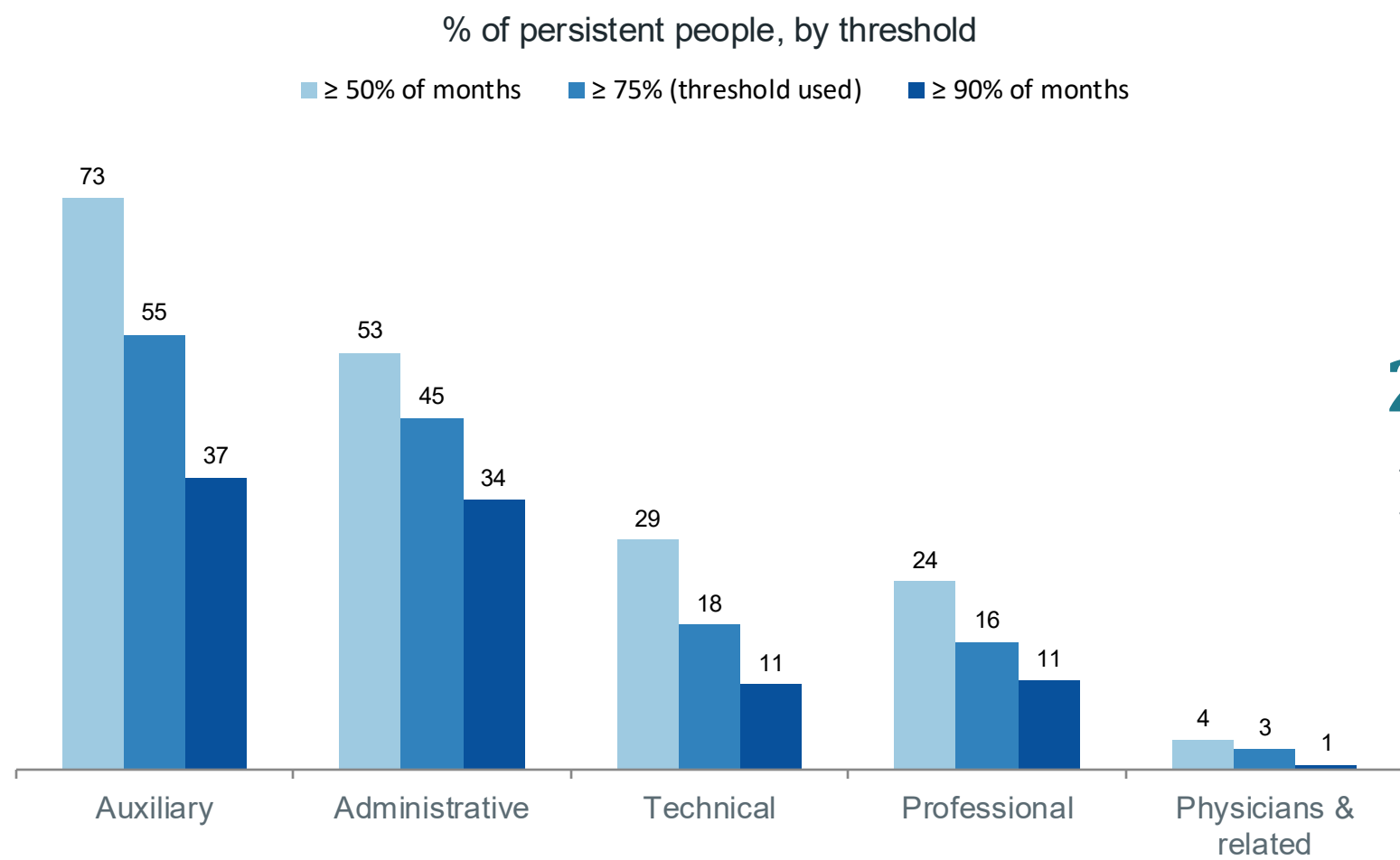
13.9–14.6%

share of gross pay

Each bar recomputes the persistence rate excluding one type of flagged case (previous slide).

Persistence: auxiliary and administrative staff lead

Across the three thresholds tested, the most and least persistent categories stay the same



55.4%

of auxiliary staff are persistent (75% threshold)

28.2% · 20,5% · 14.0%

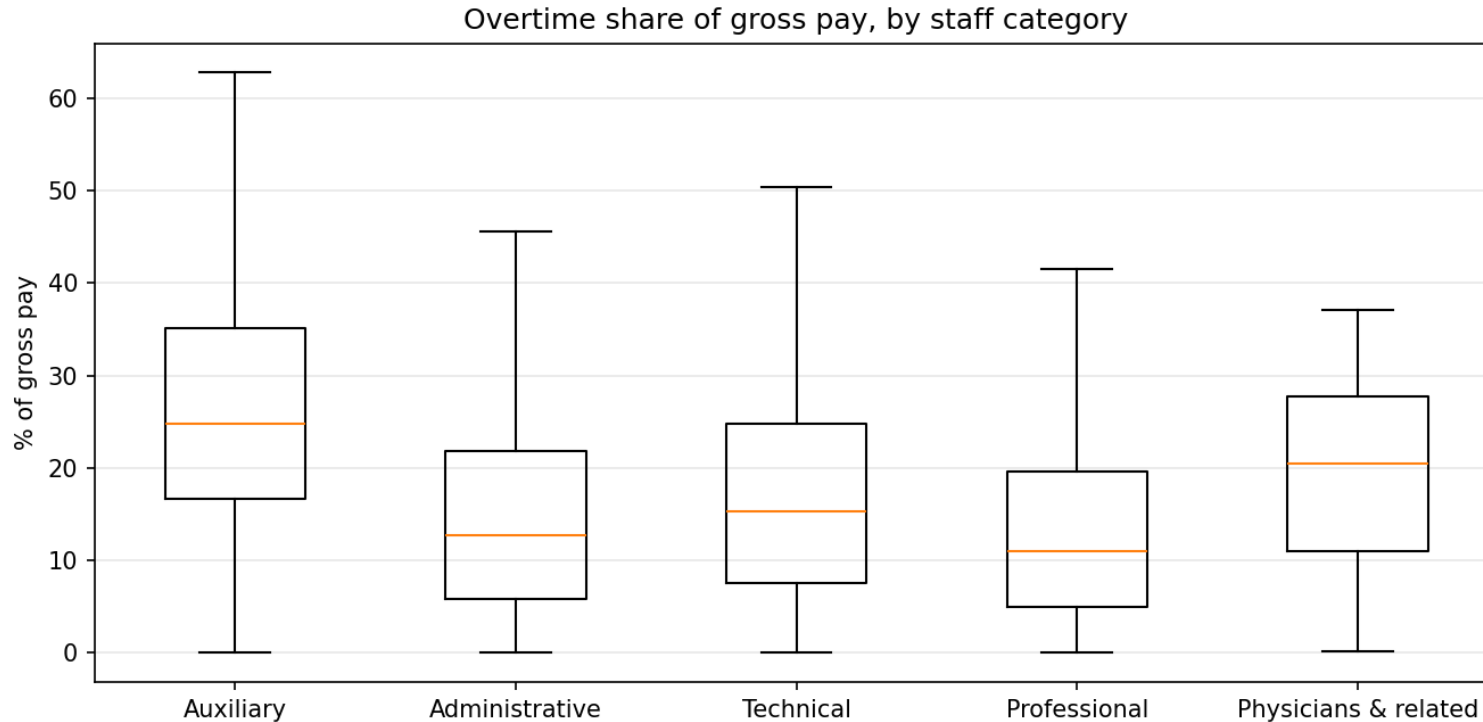
Total rate for all personnel with 50%, 75%, and 90% thresholds.

Threshold = minimum % of a person’s months with overtime to count as persistent.
Technical and professional staff are very close: at 90%, technical 10.8% vs. professional 11.3%.

Source: SSMC fixed-term staff payroll, Active Transparency portal, Sep 2024 – Aug 2026. Own analysis (analysis/EDA.ipynb).

More persistence, larger share of pay

Overtime as a share of gross pay, person-months with overtime



Pay share = overtime amount / gross pay for the month. Box: middle 50% of cases; line: median. Extreme values not shown.

24.8%

median pay share among auxiliary staff: a quarter of the month's pay

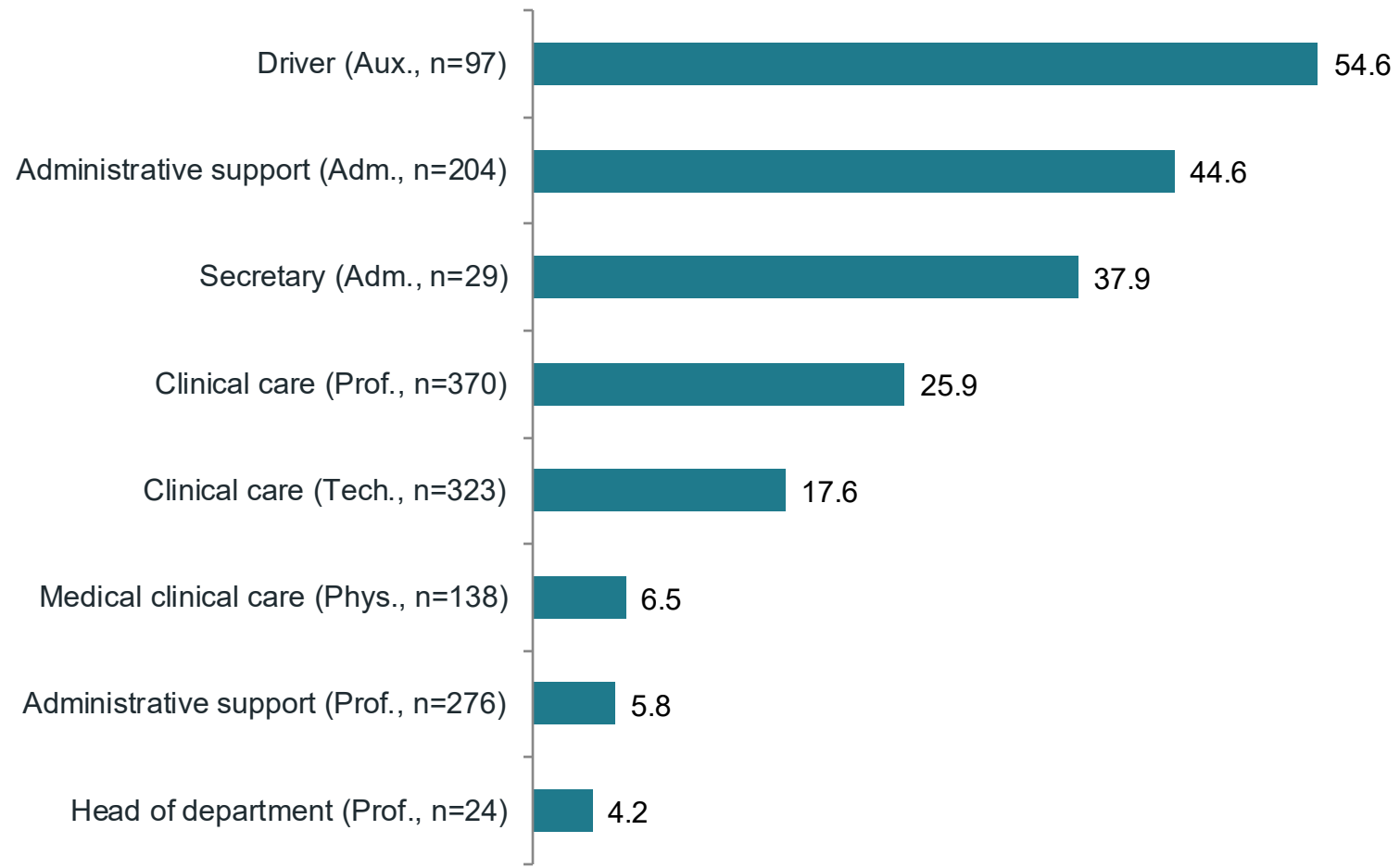
0.58

Spearman correlation between % of months with overtime and pay share (854 people; 0 = no relationship, 1 = perfect)

The relationship holds within each category (0.56–0.70), except physicians (0.18): it is not an effect of mixing groups (Simpson).

The category average hides differences between positions

% of persistent people by position · n = people with ≥ 12 months in that position (only positions with n ≥ 20)



25.9% · 5.8%

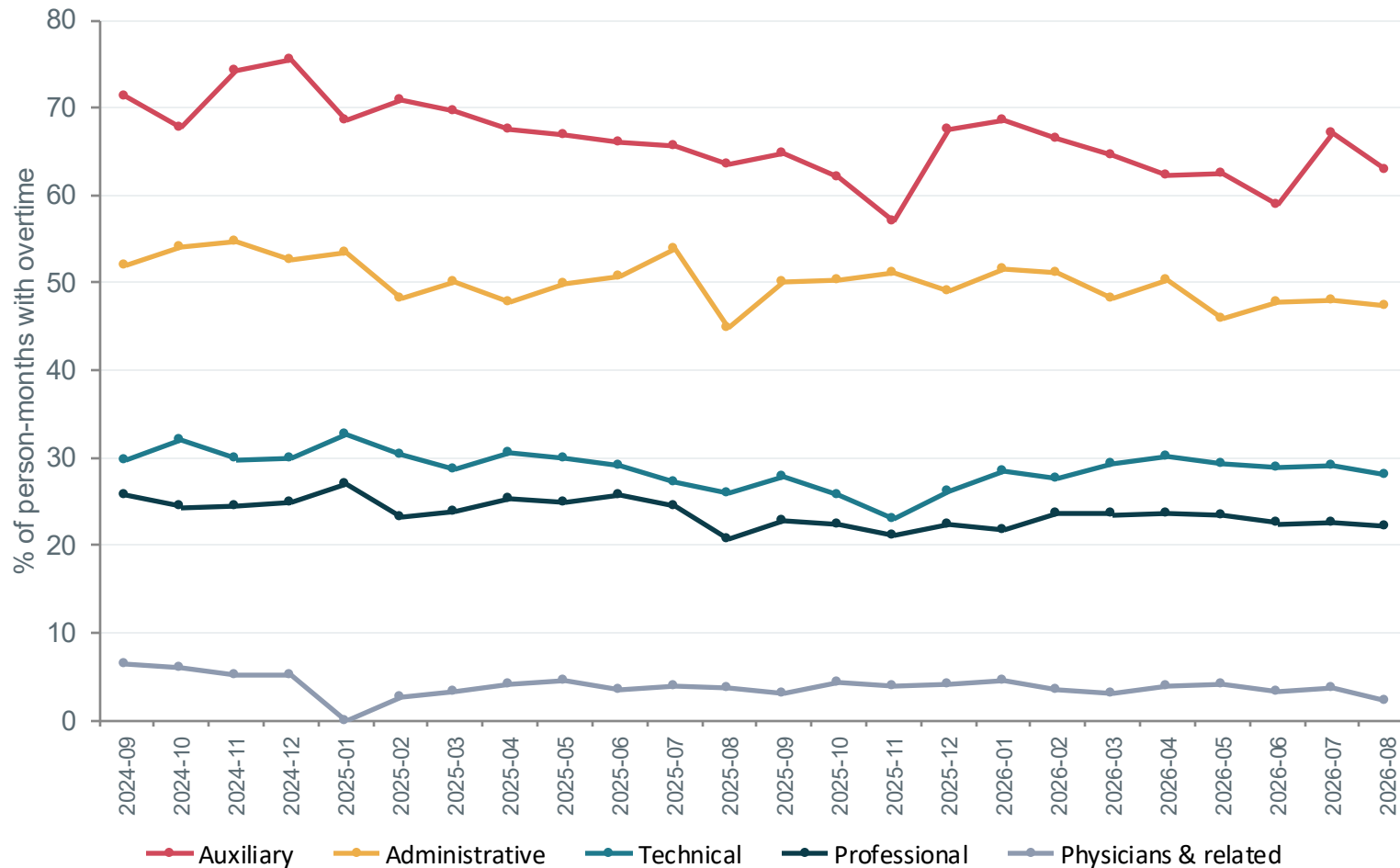
professionals: clinical care vs administrative support (category average: 16.3%)

97 of the 121 auxiliary staff (present ≥ 12 months) are drivers: the category's persistence is, in practice, that of drivers.

Physicians: pharmacist, dentistry and resident physician positions, 0%.

Monthly series and comparability

Each line: % of people in that category who received overtime in each payroll month



- The ranking of categories holds in all 24 months
- Frequency: 28.4% → 26.4% between years (Sep–Aug), in all 5 categories: not a composition effect
- August 2025: pay per hour CLP 3,623 vs 4,438–5,009; normal hours → not comparable for pay share
- January 2025: 0% of physicians with overtime

Conclusions within the evidence

Observed

1 in 5 staff members with ≥ 12 months is persistent (19.0–20.5%). It concentrates in drivers, administrative support, secretaries and clinical-care professionals, where overtime also weighs more in pay.

Interpretation

In these groups overtime appears to be a routine way of covering functions: they are the first candidates for a staffing review.

Hypotheses (not proven)

Understaffing, continuous shifts (drivers), authorization practices or accumulated back payments.

Cannot conclude

That staff is lacking, that the hours were worked, or the cause. Missing: staffing and vacancies, costs, facility, leave records.

Next steps: machine learning and dashboard

Proposals only; nothing is implemented in C1

ML task: supervised classification

- Unit: person present in both years
- Outcome: persistent in Sep 2025 – Aug 2026
- Inputs from the prior year only: category, position, working hours, overtime entitlement, % months with overtime, hours, pay share, % night hours, replacement
- No variables from the predicted year (avoids leakage)
- Alternative: clustering of usage profiles
- Uncertainty: a single time split; identification by name

Dashboard for human resources

- Decision: where to review staffing first
- Indicators: persistence, no. of persistent staff, monthly frequency, pay share
- Filters: category, position, year, month and threshold selector (50/75/90)
- Position ranking (the category average hides differences)
- Quality panel: January 2025, August 2025 (pay share), December 2025 (category)

Thank you